Accounts Role: Complete Specification, Features Matrix & SOP

Exhaustive Technical Blueprint, 12 Core Functional Modules, State Machines, Data Models & Flowcharts

Role ID: ACCOUNTS
User Profile: Accounts Officer / Cashier
Default Login: accounts@sdms.edu (acc001)

Authority: DTE Karnataka Norms
Approval Head: PRINCIPAL
Primary App: fees (SDMS Backend)

Core Security: IsAccounts Guard + Atomic Txns
Audit Trail: Automated Receipt & Reversal Logs
Format: DTE Karnataka Bilingual Standard

1. Executive Summary & Role Scope

In CampusFlow SDMS, the **ACCOUNTS** role constitutes the financial directorate of the institution. Operating under the administrative authority of the Principal and complying with Karnataka Department of Technical Education (DTE) statutory regulations, the Accounts Officer governs institutional billing, multi-seat fee structure engineering, counter cashiering, online payment verification, scholarship accounting, controlled reversals, and financial auditing.

The system enforces strict dual control (two-man rule): Accounts designs and proposes fee structures and ingests payments, while the Principal retains executive authority to approve fee structures. Payments cannot be deleted from the database—any correction requires an authorized reversal with mandatory audit reasons, ensuring 100% compliance with government accounting standards.

Institutional Governance: *Fee structures mapped to Department, Academic Year, Semester, and Seat Category automatically provision student fee ledgers across entire cohorts once approved by the Principal. Real-time balance recalculation ensures zero billing discrepancies.*

2. Complete Accounts Features Matrix (12 Modules Analyzed)

Module & UI Route	Target Area	Key Functional & Technical Capabilities
1. Accounts Dashboard /fees/dashboard	Financial Cockpit	Live KPIs: Total Collected, Today's Revenue, Total Outstanding, Concessions, Pending Verifications Count & Value. Real-time breakdown charts (by Payment Mode, Seat Type, Department). Live 15-entry transaction ledger with instant receipt modal and inline reversal.
2. Fee Structure Engineering /fees/create	Curriculum Billing	Configure seat-type fee structures (GM, SNQ, Management, Lateral, NRI). Encapsulates 14 authentic Karnataka DTE breakdown components. Live DTE receipt preview recalculates real-time sums during editing.
3. Principal Approval Pipeline /fees/detail/:id	Governance & Dual Control	Accounts submits draft (pending) → Principal reviews → Upon approval, system atomically queries matching cohort students and provisions StudentFee ledger records with automated push notifications.
4. Student Fee Ledger /fees/students	Cohort Directory	Multi-filter student fee directory (by USN, Name, Department, Semester, Status). Computes Total Assigned, Net Payable, Total Paid, and Outstanding Balance in real time. Actions: View Statement, Apply Concession, Direct Pay.
5. Counter Payment Recording /fees/record-payment	Cashiering & Collection	In-person fee collection. Auto-fetches unrecorded student dues upon selection. Supports Cash, UPI/QR, Bank Transfer, DD, and Cheque. Component allocation ('ALL' or specific item). Generates official DTE receipt instantly.
6. Payment Verification Queue /fees/student-payments	Proof Verification	Review queue for student-uploaded transaction slips and UPI screenshots. High-resolution document viewer with pan/zoom. Accounts can Verify (issues official DTE receipt) or Reject with mandatory audit remark.
7. Uploaded Proofs Archive /fees/uploaded-files	Document Repository	Centralized visual archive of all student payment counterfoils, challans, and transaction screenshots. Grid & list views, filter by Department, Semester, Academic Year, and Verification Status.
8. Controlled Payment Reversal reverse() API	Audit & Refund Control	Enterprise reversal workflow for bounced cheques or erroneous entries. Requires mandatory justification. Flags record as is_reversed=True, restores student balance atomically, and notifies student.
9. Concessions & Scholarships apply-concession	Welfare & Fee Waivers	Deducts government scholarships (SSP, NSP, EWS, Merit) from gross fees. Tracks approving officer and timestamp. Automatically recalculates balance and updates student payment standing.
10. Official DTE Receipt Engine DteFeeReceiptWidget	Statutory Compliance	Pixel-perfect replication of Karnataka DTE receipt booklet format. Features bilingual title, unique sequential receipt & booklet numbering, 14 standard component items, and automatic Indian currency words converter.
11. Academic Year Completion academic-year-summary	Annual Standing	Annualized financial tracker computing student standing across both odd & even semesters. Renders percentage completion progress bar and status badge (COMPLETED / PARTIALLY_COMPLETED / NOT_COMPLETED).
12. Financial Reports & Exports /fees/reports	Auditing & Reconciliation	4-tab analytical reporting suite: (1) Daily Collections, (2) Defaulters List, (3) Department Breakdown, (4) Concessions Audit. Instant 1-click CSV download for treasury and auditor review.

3. Detailed Module Specifications & Technical Workings

The 14 Authentic Karnataka DTE Fee Breakdown Components

In compliance with Government of Karnataka Department of Technical Education norms, CampusFlow structures each academic year's fee schedule into 14 distinct statutory heads. When an Accounts Officer creates a fee structure, these components are populated and validated:

1. Registration Fee	5. Library Fee	9. ID Card Fee	13. NSS Fee
2. Tuition Fee	6. Magazine Fee	10. Student Welfare Fee	14. Other / Misc Fee
3. Sports Fee	7. Library Tax	11. Teacher's Welfare Fee	**Total Fee:** Sum of items
4. Union Fee	8. Laboratory Fee	12. Development Fee	**Governed By:** DTE Karnataka

Dual Payment Channels: Direct Counter Collection vs. Online Student Uploads

CampusFlow supports two parallel payment ingestion pathways designed to accommodate both offline counter cashiering and modern digital payments:

• **Track A — In-Person Counter Payments (/fees/record-payment):** Student visits accounts window. Cashier selects student → dues auto-load → enters payment mode (Cash, UPI, Cheque, DD, Bank Transfer) → system validates uniqueness of reference number → executes atomic database commit → immediately generates official DTE receipt → marks payment verified=True.

• **Track B — Student Online Self-Submission (/fee-upload):** Student transfers fees via bank or UPI QR code → uploads screenshot or counterfoil → payment record created with verified=False → queued in Accounts Verification Queue → Accounts inspects document → Verifies (issues receipt) or Rejects (with mandatory feedback).

Official DTE Bilingual Receipt Engine & Number-to-Words Converter

The DTE Receipt widget (dte_fee_receipt_widget.dart) meticulously mirrors the official physical receipt booklet issued across Karnataka government colleges. Key elements include:

- **Bilingual Government Banner:** Karnataka Government seal, Department of Technical Education, and institution header.
- **Unique Numbering:** Booklet Series (e.g. 10495...) and sequential 7-digit receipt identifier (e.g. 0821183).
- **Indian Currency Words Algorithm:** Custom utility converts numbers into authentic Indian currency prose (e.g. ■*32,932* → *'Thirty Two Thousand Nine Hundred and Thirty Two Rupees Only'*).
- **Official Signatures & Seals:** Pre-configured signature blocks for Cashier and Accounts Officer with print/PDF export readiness.

Cross-Module System Integrations (Corrections, Grievances, Announcements)

The ACCOUNTS role is deeply woven into institutional administration across multiple SDMS applications:

- **Student Data Corrections (backend/corrections/views.py):** Accounts is automatically routed any correction request under the FEES category (e.g., student quota updates or fee category revisions).
- **Grievance Redressal (backend/grievances/views.py):** Fee-related grievances submitted by students or parents are dynamically assigned to the Accounts Officer for investigation and response.
- **Principal Directives & Announcements (backend/announcements/):** Accounts receives real-time official directives, executive budget orders, and fee structure approval notifications directly in the notifications hub.
- **Library Fine Integration (backend/library/signals.py):** Overdue book fines recorded by the librarian trigger automated signals notifying Accounts of pending student dues.

4. System Flowcharts & Process State Machines

The visual architectures below outline the life cycles of fee structures, payment verification, and ledger reconciliation.

Flowchart 1: End-to-End Institutional Fee Lifecycle

Step / Actor	Operational Process Flow	Database & State Transition
1. Accounts Officer	Drafts fee structure for Dept, Sem, Academic Year & Seat Category. Adds 14 DTE components. Submits for review.	`FeeStructure(approval_status='pending', is_active=False).` `Notification sent to Principal.`
2. Principal	Reviews total fee, component breakdown, and regulatory compliance. Approves or Rejects with remark.	`If Approved: approval_status='approved', is_active=True.` `If Rejected: rejection_remark='...'.`
3. Automated Engine	System queries active students matching Dept, Sem, Academic Year, and Seat Type. Provisions student ledgers.	`Bulk creates StudentFee records. Net payable computed. Dispatches push notification to students.`
4. Ingestion Channel	**Path A (Counter):** Cashier enters payment directly. **Path B (Online):** Student pays and uploads slip.	`Path A: FeePayment(verified=True).` `Path B: FeePayment(verified=False).`
5. Verification & Receipt	Accounts verifies Path B proof or completes Path A cashiering. System issues official DTE receipt.	`Generates FeeReceipt record.` `Updates StudentFee.status → 'partial' or 'paid'.`
6. Audit & Reporting	Accounts monitors daily collections, exports defaulter reports, and reconciles treasury accounts.	`Data exported via FeeReportsView as downloadable CSV spreadsheets.`

Flowchart 2: Payment Verification & Reversal State Machine

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State Progression:
[Student Uploads Proof Slip] → PENDING VERIFICATION (verified=False, is_reversed=False)
    ■→ Accounts Action: REJECT → REJECTED (Remark recorded, student notified to re-upload)
    ■→ Accounts Action: VERIFY → VERIFIED & ACTIVE (verified=True, Official DTE Receipt generated)

[Counter Direct Cashiering] → VERIFIED & ACTIVE (verified=True, DTE Receipt issued immediately)
    ■→ Authorized Action: REVERSE → REVERSED (Reason logged, balance restored, receipt invalidated)
```

5. Database Schema & Architecture (fees App)

The backend architecture in backend/fees/models.py comprises 5 core normalized relational models:

Model Name	Key Attributes & Data Fields	Constraints & Technical Logic
FeeStructure fee_structure	department, academic_year, semester, seat_type, title, description, is_active, approval_status, approved_by, approved_on, created_by, rejection_remark	Unique on (dept, academic_year, sem, seat_type, title). Total calculated as dynamic sum of child components.
FeeComponent fee_component	fee_structure (FK), name (max 80 chars), amount (decimal 12,2)	Cascade delete with parent structure. Unique constraint on (fee_structure, name).
StudentFee student_fee	student (FK), fee_structure (FK), due_date, status (pending/partial/paid), concession_amount, concession_reason, concession_approved_by, concession_approved_on	Unique on (student, fee_structure). Balance = (Total - Concession) - Paid. Auto-calculates status on payment/reversal.
FeePayment fee_payment	student_fee (FK), amount, mode (cash/upi/bank/dd/cheque/online), paid_on, reference_no, upi_txn_id, upi_screenshot, bank_reference_no, bank_statement_file, entered_by, verified, remark, is_reversed, reversed_at, reversed_by, reversal_reason	Clean method enforces duplicate reference checks across active payments. Reversal tracks actor and reason without hard deletion.
FeeReceipt fee_receipt	payment (OneToOne), receipt_no, issued_on, verified_by	Receipt numbers formatted via REC-YYYYMMDDHHMMSSffffff or sequential counter. Protected on delete.

6. Standard Operating Procedures (Accounts Staff SOP)

SOP 1: Creating and Submitting a New Fee Structure

1. Log in with Accounts credentials (accounts@sdms.edu) and navigate to **Fee Structure** → **Create Structure** (/fees/create).
2. Select Department (e.g. CSE), Academic Year (2025-2026), Semester (Sem 3), and Seat Quota (General Merit).
3. Enter an official title (e.g. *'B.E. CSE 3rd Sem GM Fee Structure'*).
4. Populate the 14 standard DTE fee components. The live DTE receipt preview dynamically computes the gross fee.
5. Click **Submit for Principal Approval**. The structure is saved as 'Pending Principal Approval'. The Principal receives an instant directive notification.

SOP 2: Direct Counter Payment Recording (Cashier Window)

1. Navigate to **Record Payment** (/fees/record-payment).
2. In the Search Student bar, enter the student's USN or name. Click to select.
3. The system automatically populates the student profile, fee breakdown, remaining balance, and the Academic Year Completion card.
4. Select Payment Mode (Cash, UPI, Bank Transfer, DD, Cheque). If UPI or Bank, enter the transaction/reference number.
5. If paying for a specific fee head (e.g. Tuition Fee only), pick that component; otherwise leave as 'ALL'.
6. Click **Record Payment & Generate Official Receipt**. The official DTE receipt dialog appears with instant print and PDF options.

SOP 3: Verifying or Rejecting Student Online Payment Uploads

1. Navigate to **Verify Payments** (/fees/student-payments).
2. Filter by status 'Pending'. Inspect the student USN, claimed amount, and click the document thumbnail to launch the high-res proof viewer.
3. Cross-reference the transaction ID against institutional bank credit advice. If valid, click **Verify & Issue Receipt**. If invalid or blurred, click **Reject** and supply a detailed reason.

SOP 4: Executing an Authorized Payment Reversal

1. From the **Accounts Dashboard** (/fees/dashboard), locate the transaction under **Recent Transactions**.
2. Click the red **Reverse** button.
3. Provide an authorized justification (e.g. *'Cheque #402910 bounced due to insufficient funds'*).
4. Confirm reversal. The payment status changes to REVERSED, the student's balance is automatically restored, and an audit notification is dispatched.

SOP 5: Applying Fee Concessions & Government Scholarships

1. Open **Student Fees** (/fees/students) and locate the student by USN.
2. Click the **Apply Concession** action icon.
3. Enter the concession amount (must not exceed gross fee) and the scholarship scheme name (e.g. *'SSP Post-Matric Scholarship'* or *'Sports Merit Concession'*).
4. Click Save. The net payable balance recalculates immediately and updates student records across all portals.

SOP 6: Generating & Exporting Financial Reports

1. Open **Financial Reports** (/fees/reports).
2. Select desired tab: **Daily Collections**, **Defaulters List**, **Department Summary**, or **Concessions**.
3. Review analytics or click **Export CSV** to download the formatted audit spreadsheet.

7. Security, Compliance & Audit Controls

- **Role-Based Access Control (RBAC):** Direct payment recording, manual verification, and reversal are protected by the IsAccounts permission class. Unauthorized roles receive strict 403 Forbidden responses.
- **Atomic Database Transactions (@transaction.atomic):** Payment creation, receipt numbering, status recalculation, and notification dispatch are executed inside atomic blocks. If any step fails, the entire transaction rolls back cleanly.
- **Anti-Duplicate Safeguards:** All transaction references are cross-checked across the database to prevent duplicate payment claims.
- **Immutability of Financial History:** Payment records are never hard-deleted; reversals maintain a full audit trail including timestamp and approving user identity.

End of Specification Document — CampusFlow SDMS Accounts Role & Financial Directorate